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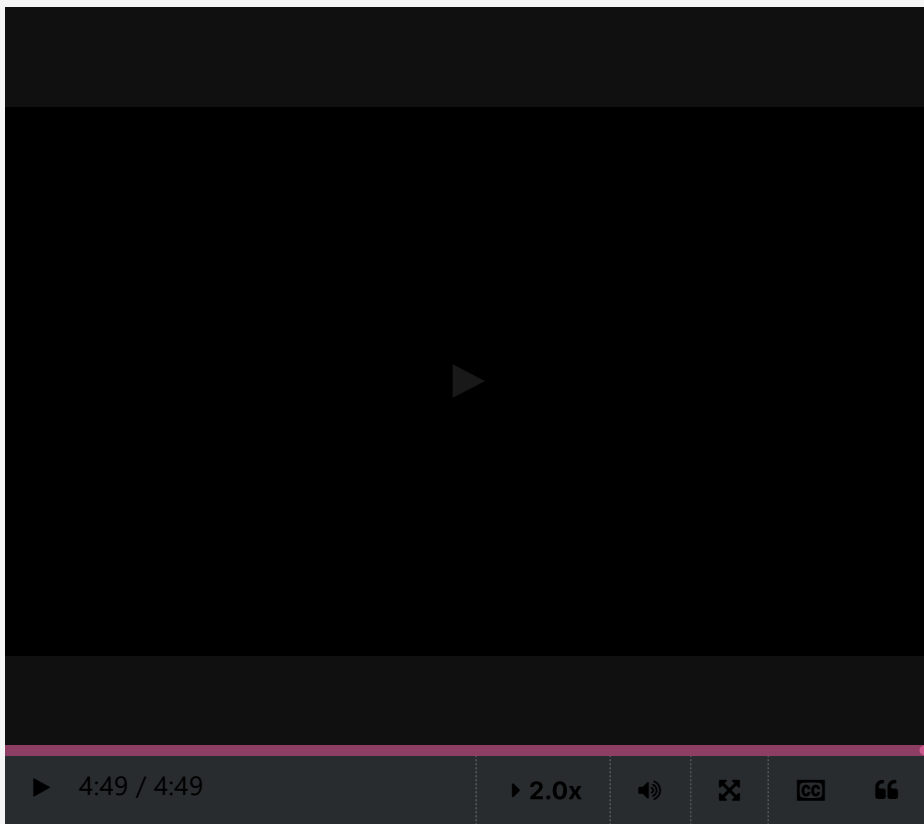
Complexity cannot be computed! But...

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XPERT

Problem: Complexity is not computable. Does it mean that it is useless?

Kolmogorov complexity cannot be computed!



complexity.
Conclusion:
Kolmogorov complexity is an ideal notion that can be approximated, but that is not computable. It is nevertheless an enlightening concept for many developments in **Artificial Intelligence.**

~ .

program uses `cc` to

```
ge(2**i):  
m)[2:]  
> n:    return s
```

icient: it starts over the
nd again until it reaches
re complex than n .

Multiple Choice

1/1 point (ungraded)

Can we compute the complexity $C(s)$ of an object s on a specific universal Turing machine if one is sure that $C(s) < 20$ bits?

☐ Yes.

☐ Possibly.

☒ No.



Answer
Correct:
We restrict ourselves to one specific universal Turing machine U . Why not try all programs of length < 20 bits and see which of them computes s ? The length of the shortest among them would be $C(s)$. Unfortunately, there is no guarantee that all these programs will stop after a finite amount of time when executed on U . If one puts an upper bound to execution time, one may miss the true value of $C(s)$. So "No" is an acceptable answer.

If we are lucky, we may find that a program outputs s on U and while all shorter programs don't. This would provide the true value of $C(s)$ on U . But there is of course no guarantee that it would work for any s . So "Possibly" is a correct answer.

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You have used 1 of 1 attempt

✓ Correct (1/1 point)

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